

2(a)	1 genetic / genotype / alleles or example ; mutation / crossing over / independent assortment / random fertilisation / random mating 2 environmental or example ; climate / disease / food availability / soil pH / selection pressure 3 combination / interaction, of genes <u>and</u> environment or example ; gene (expression) is modified by environment	3
2(b)(i)	any three from: 1 formation of new species from, pre-existing species / common ancestor ; 2 change in, characteristics / phenotypes ; 3 over, time / generations ; 4 <u>natural selection</u> ; 5 changes in, gene pools / allele frequencies or genetic drift ;	3

Question	Answer	Marks
2(b)(ii)	any three from: 1 compare sequences (of DNA) ; 2 the more similar the sequences the more, closely related the species / recent the common ancestor ; ora 3 over time mutations accumulate or less time fewer mutations occurred ; 4 <i>ref. to</i> bioinformatics / software / database ; 5 AVP ; e.g. molecular clock / constant rate of mutation e.g. mtDNA has no crossing over e.g. the percentage similarity can be used to build a phylogenetic tree	3